PHYC10009 Physics Foundations

Modern Physics: Wave nature of matter
Week 10 Lecture 2

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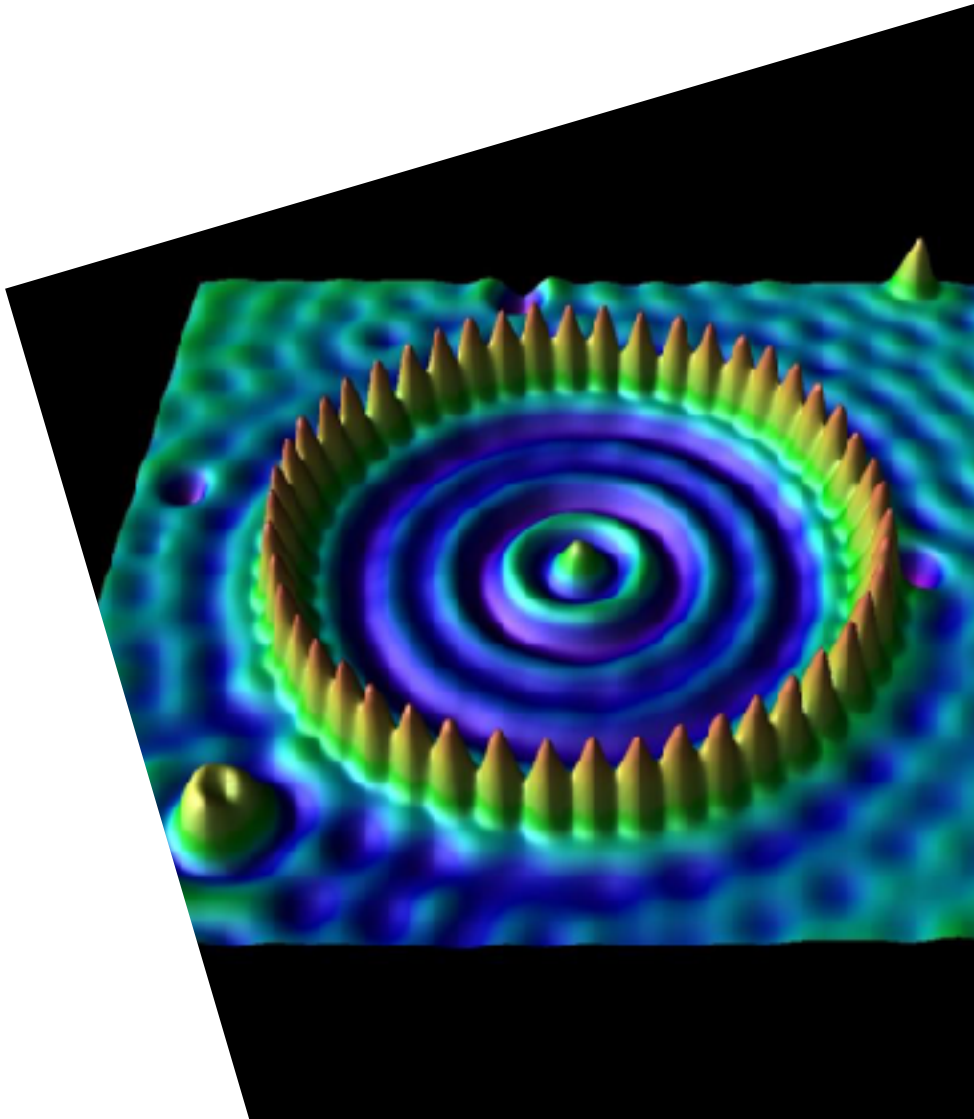
School of Physics
University of Melbourne



This lecture

27.4 Photon mass and momentum

27.8 Wave nature of matter

28.3 Heisenberg Uncertainty Principle

27.12 Atomic structure/Bohr model

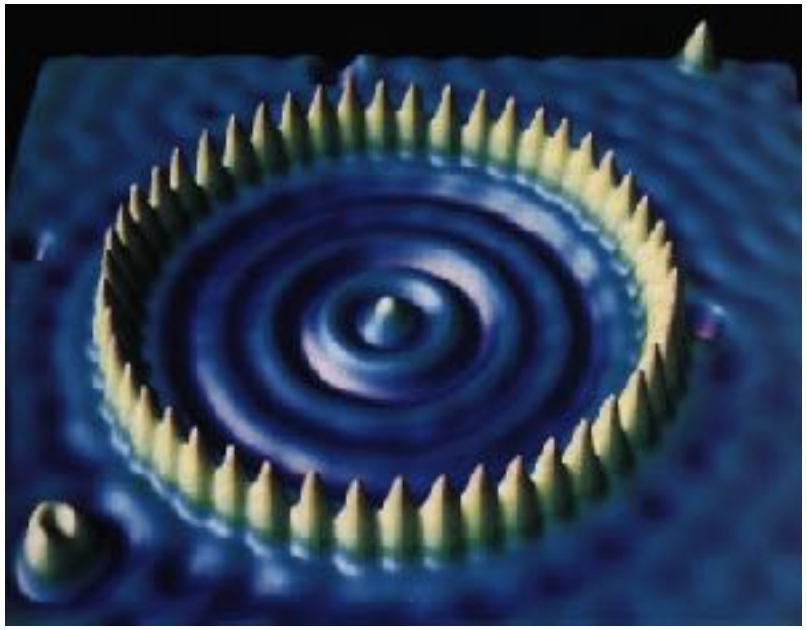
Matter Waves

We've seen that ElectroMagnetic "waves" behave like particles.

The particle-like property of waves, their energy, depends on the wave-like nature, ie Their frequency.

It turns out that particles also behave like waves.

The wave-like nature of particles, their wavelength, λ depends on their particle-like property, their momentum.



$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Where:
λ= wavelength
h = Plancks Constant
p = Momentum
m = mass
v = velocity

Matter Waves

The "wave-length", λ, is called the "deBroglie" wavelength of a particle

Question: What is the deBroglie Wavelength of an electron with a kinetic Energy of 1 eV?

Kinetic Energy K = ½ m v^2 = 1.0 eV = 1.6 x10^{-19}J

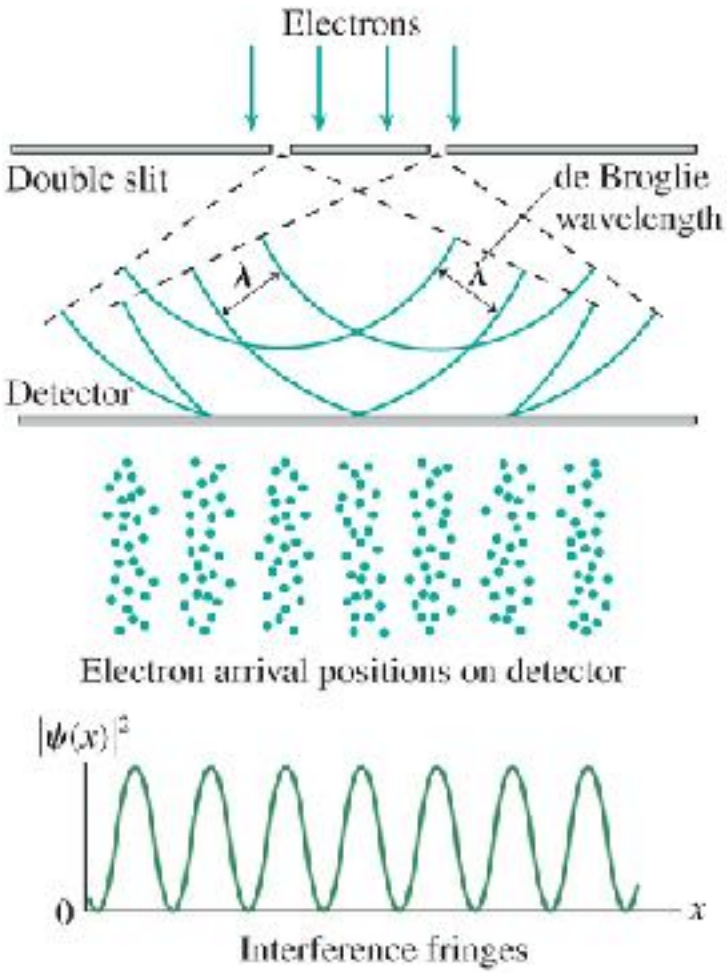
$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} J}{9.1 \times 10^{-31} kg}} = 5.8 \times 10^5 \quad \text{ms}^{-1}$$

$$\lambda = \frac{h}{mv} = \frac{6.6x10^{-34}}{9.1 \times 10^{-31} \times 5.9 \times 10^5} = 1.2\text{x}10^{-9}\text{m}$$

This distance is approximately the separation between atoms

Matter Waves

The wave-like nature of particles is conclusively demonstrated through Interference experiments.



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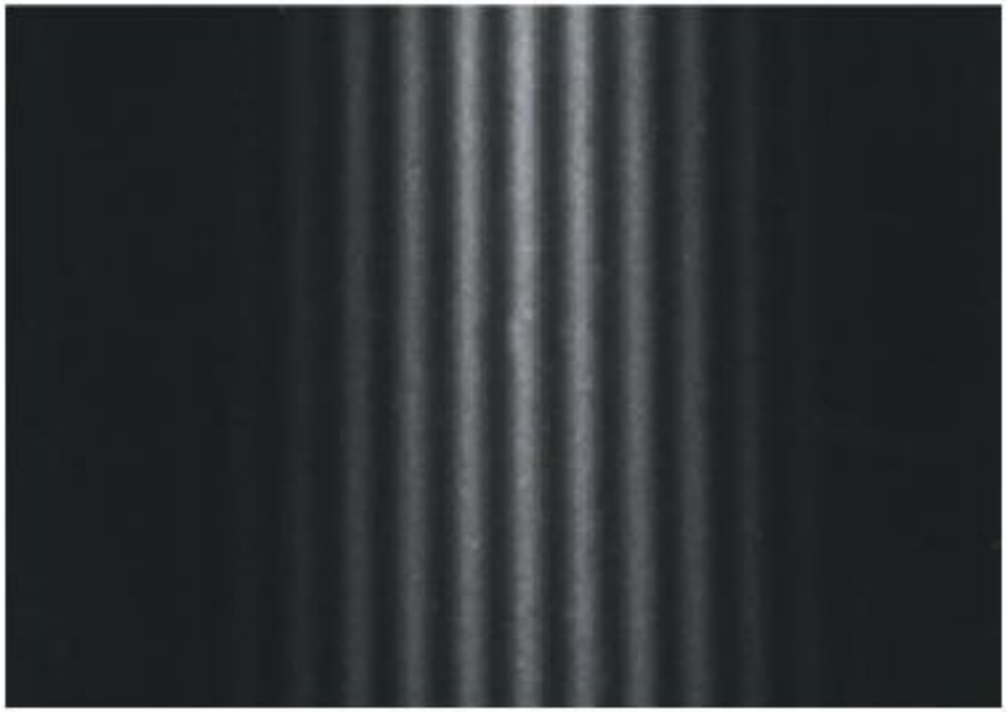

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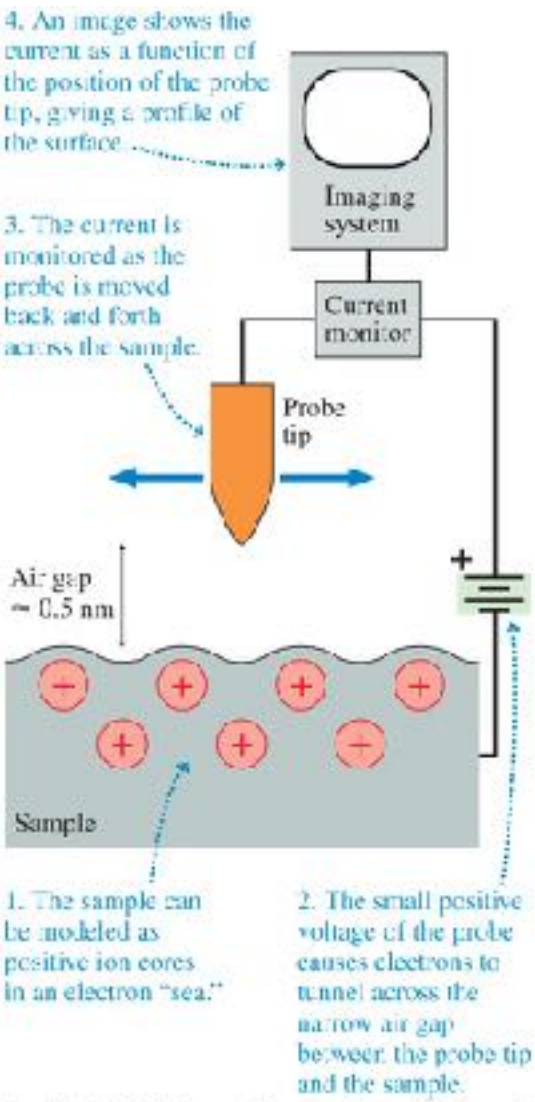
Matter waves

And through sub-atomic precision measurement with Scanning Tunneling Microscopes.



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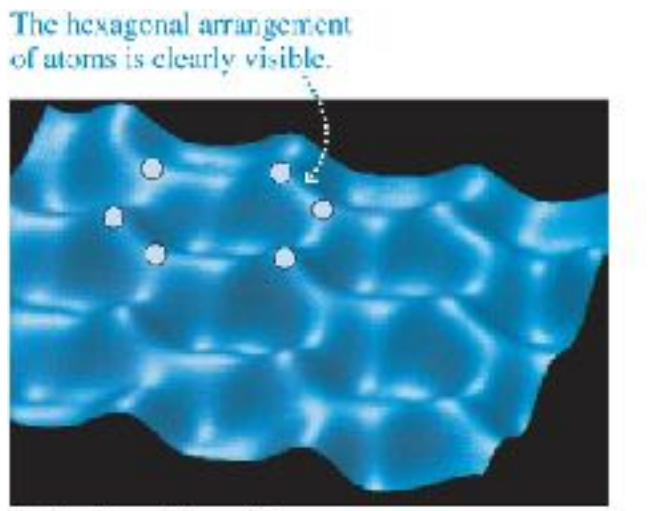

(a) Surface of graphite

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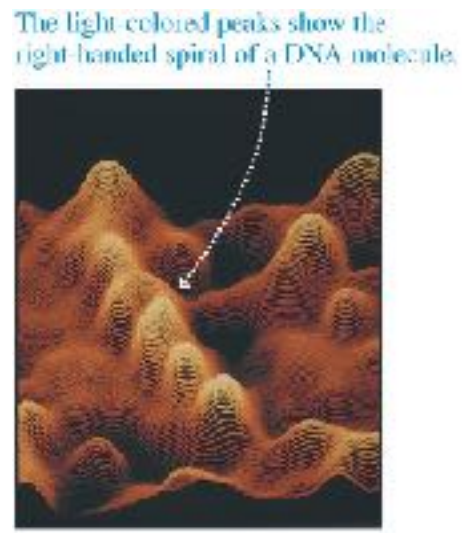

(b) DNA molecule

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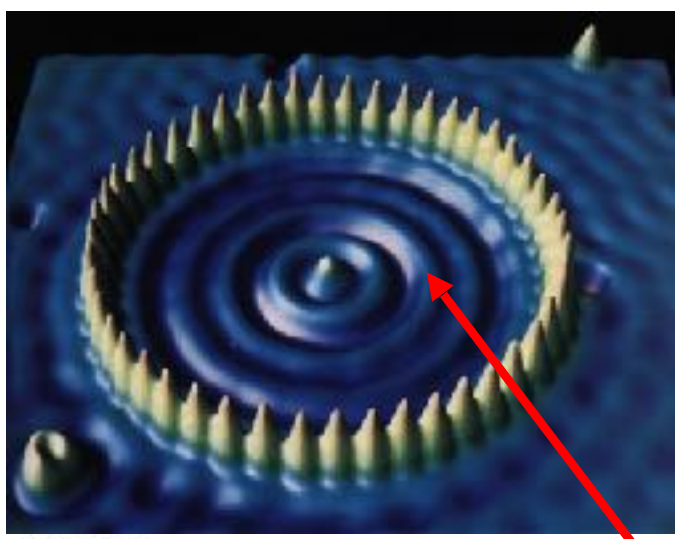
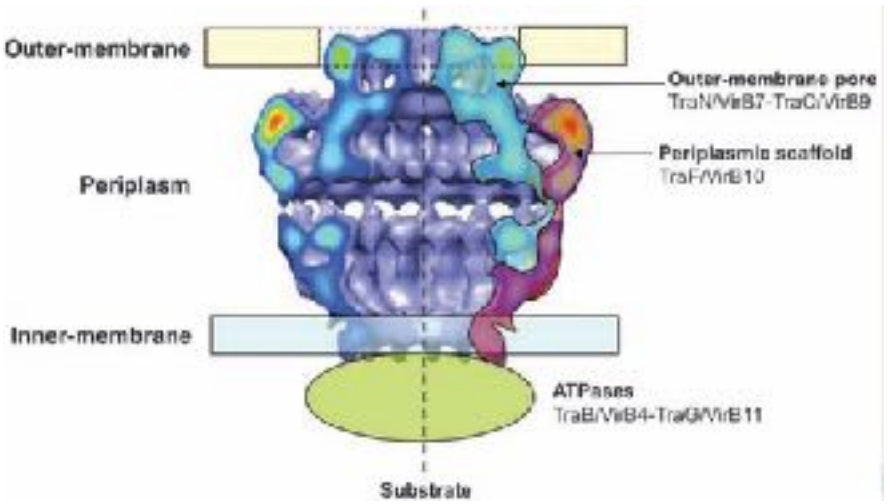


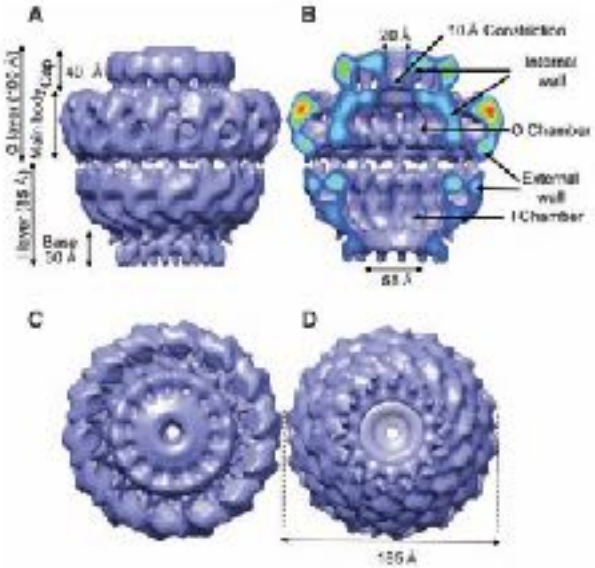
Ring of Chromium Atoms enclosing Electrons on a Metal surface

Rings of electron probability density

Matter waves at the Biomolecular scale

Biological Molecules employ proteins to construct nano-machines vital for life.

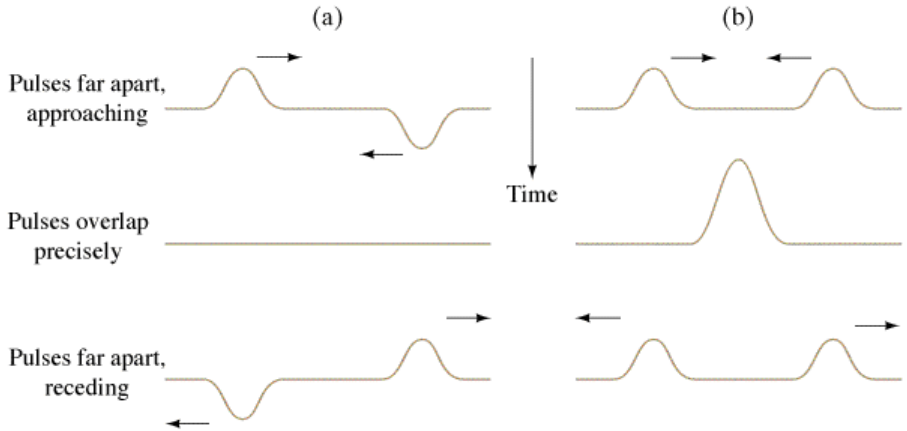




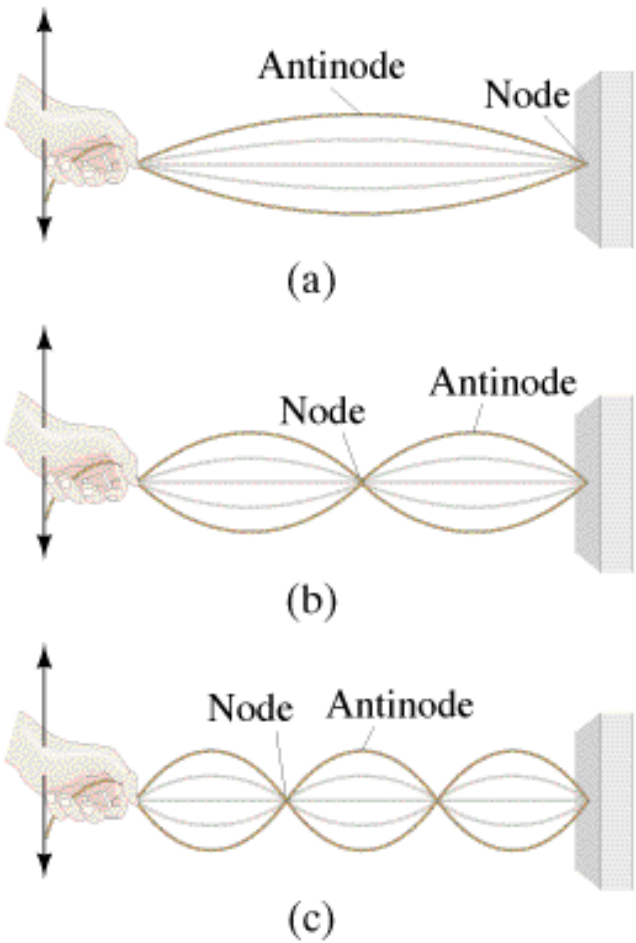
Channel constructed by plasmids to penetrate cell membrane and infect neighbouring cells.
Color shows the electron density generated by the proteins.
(Fronzes et al. Science, V 323, P266, (2009))

Reminder

Principle of Super Position: The amplitude from the combined effect of more than one wave is the sum of the amplitudes of each wave:

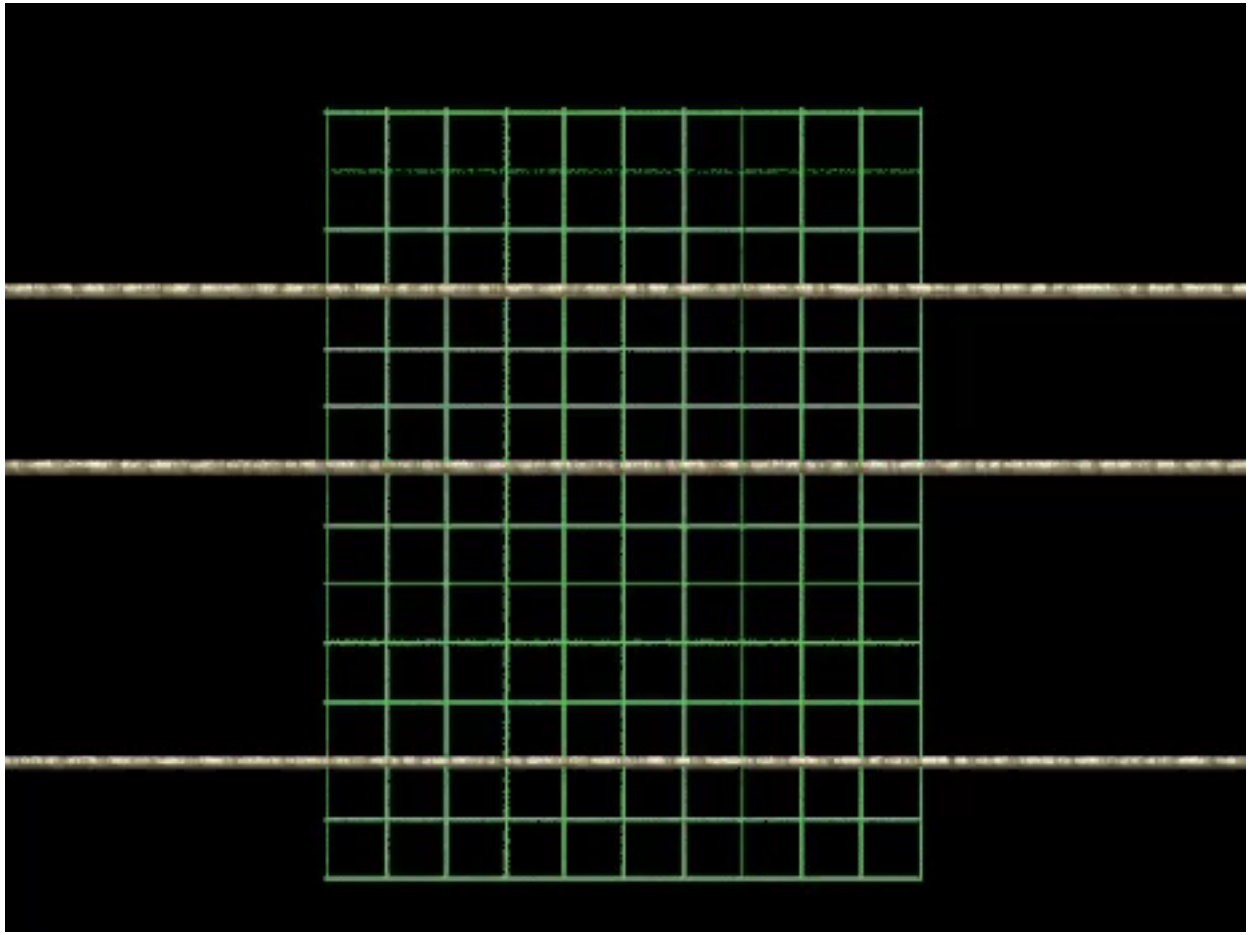


Standing Waves



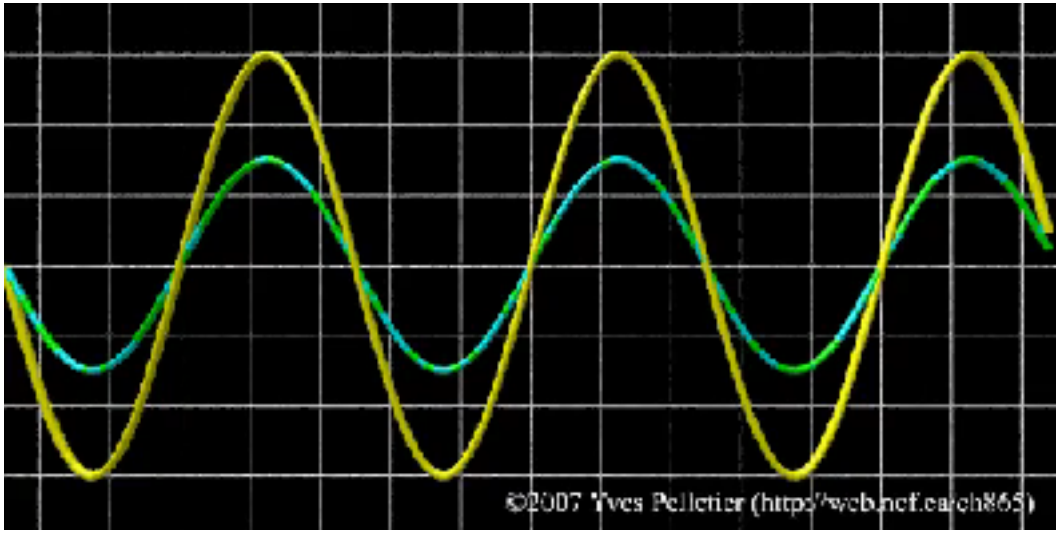
Principle of Super Position

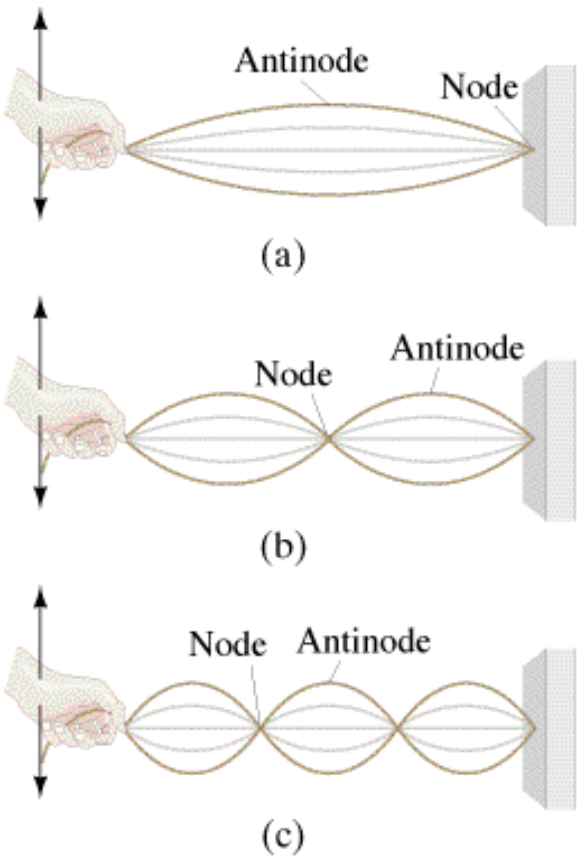
Pulses interfering



Interference: Standing waves (16.1/16.2)

If wave energy is conserved at reflection in a cavity and the size of the Container forms nodes at the ends of cavity, a "Standing Wave" is formed





- Oscillate string with a fixed termination – wave propagation
- Interference of waves traveling back and forth along the string
- At appropriate frequencies, the two waves interfere to form a large-amplitude **standing wave**
- An example of **Resonance**

Energy is Quantised

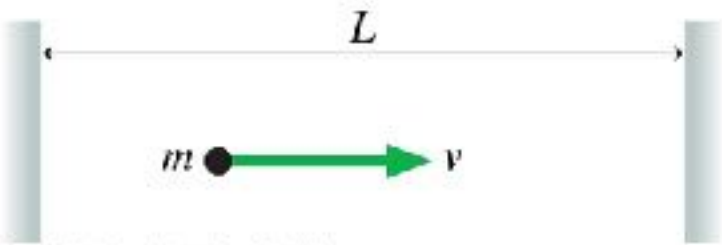
We've seen that particles have specific wavelengths related to their momentum

$$\lambda = \frac{h}{p}$$

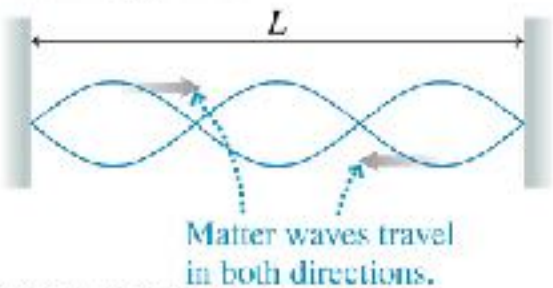
λ = wavelength
h = Plancks'Constant
p = momentum

When particles are confined to specific regions of space, their wave-nature gives rise to standing waves.

(a) A classical particle of mass m bounces back and forth between two boundaries.



(b) Matter waves moving in opposite directions create standing waves.



$$\lambda_n = \frac{2L}{n} \quad \Rightarrow \quad \frac{h}{p} = \frac{2L}{n} \quad \Rightarrow \quad p_n = n\left(\frac{h}{2L}\right)$$

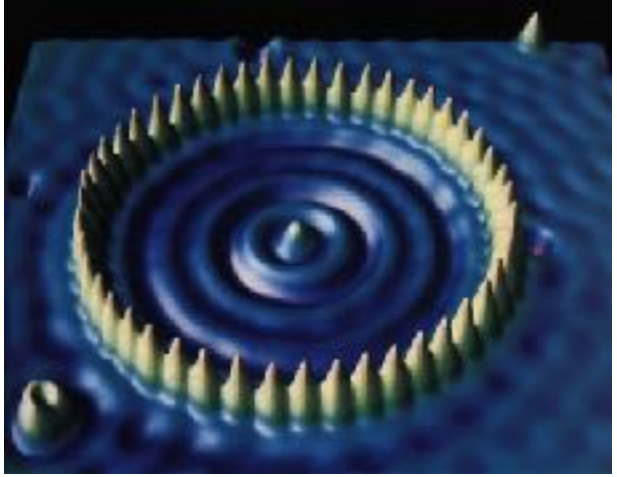
Now $E = \frac{p^2}{2m}$

So that Energy is Quantised

$$E_n = \frac{1}{2m}\left(\frac{hn}{2L}\right)^2 = \frac{h^2}{8mL^2}n^2$$

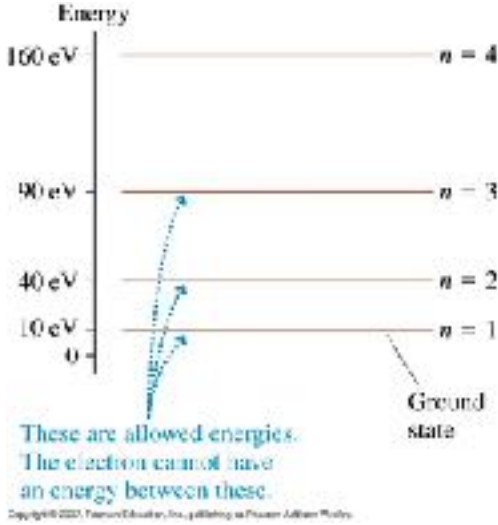
Energy Levels

At Atomic sizes the wavelengths of electrons are comparable to the size of their confining regions. Hence they exhibit standing waves.

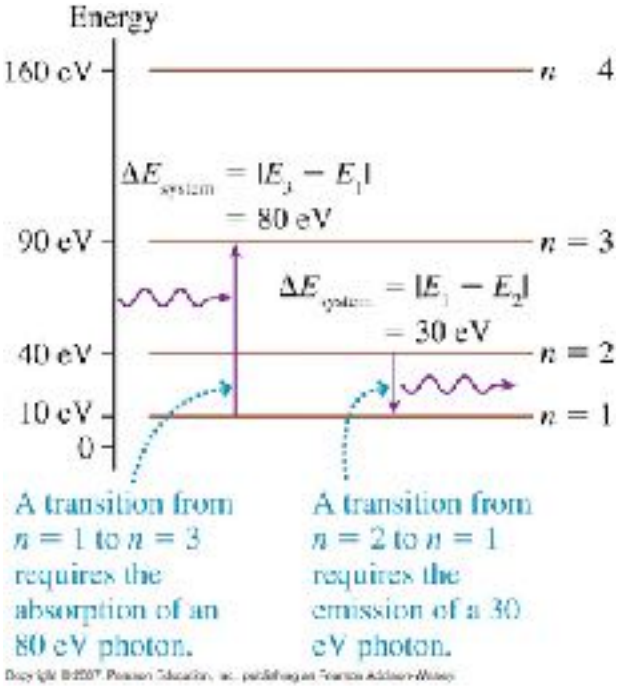


Because of this quantum systems can **only** exist at specific energy levels

An energy level diagram shows the allowed energy levels



Atomic systems make transitions between energy levels by absorbing or emitting photons

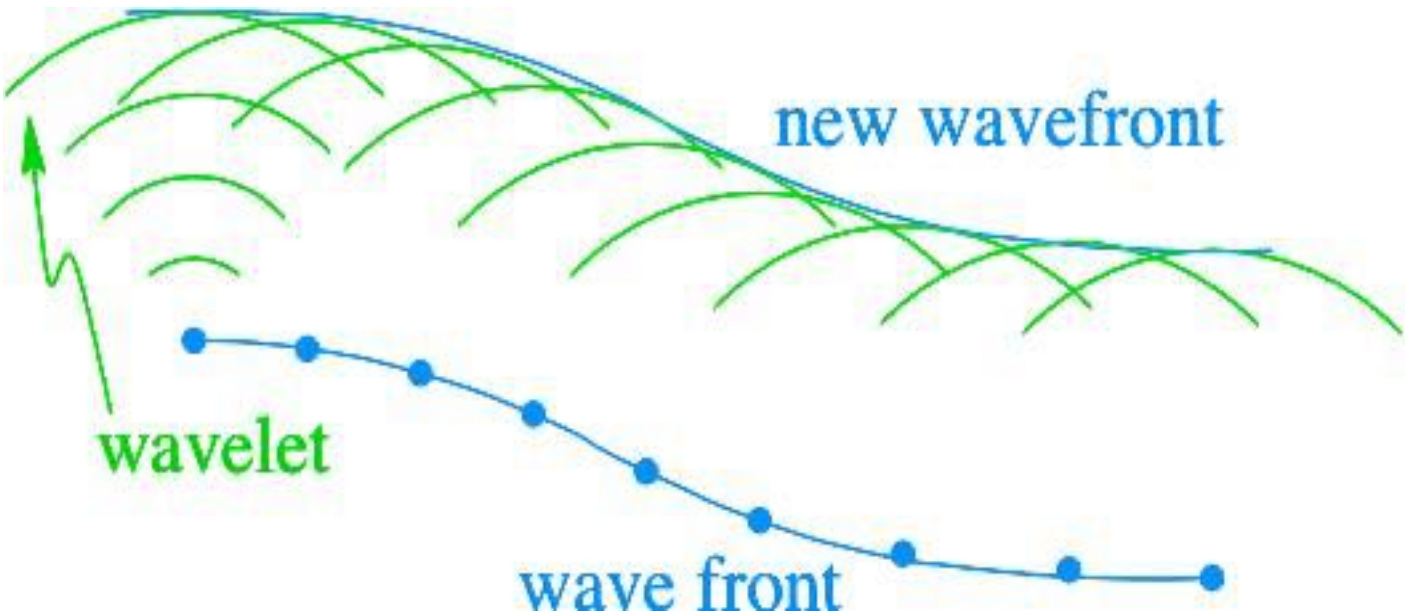


27.4 Photons: energy, mass, momentum

- Photons travel at the speed of light ☺
- Momentum $p = mv$ but at relativistic speeds $p = \dfrac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}$. Oops… $v = c$?
- Mass must be zero! But rest energy $E = mc^2 = 0$ (OK, photons never at rest)
- *All* photon energy is kinetic energy: $\mathrm{KE} = E = hf$
- Momentum from relativity $E^2 = p^2c^2 + m^2c^4$ but $m = 0$ so $p = \dfrac{E}{c} = \dfrac{hf}{c} = \dfrac{h}{\lambda}$
- Photon momentum: $p = \dfrac{h}{\lambda}$

Huygens' Principle

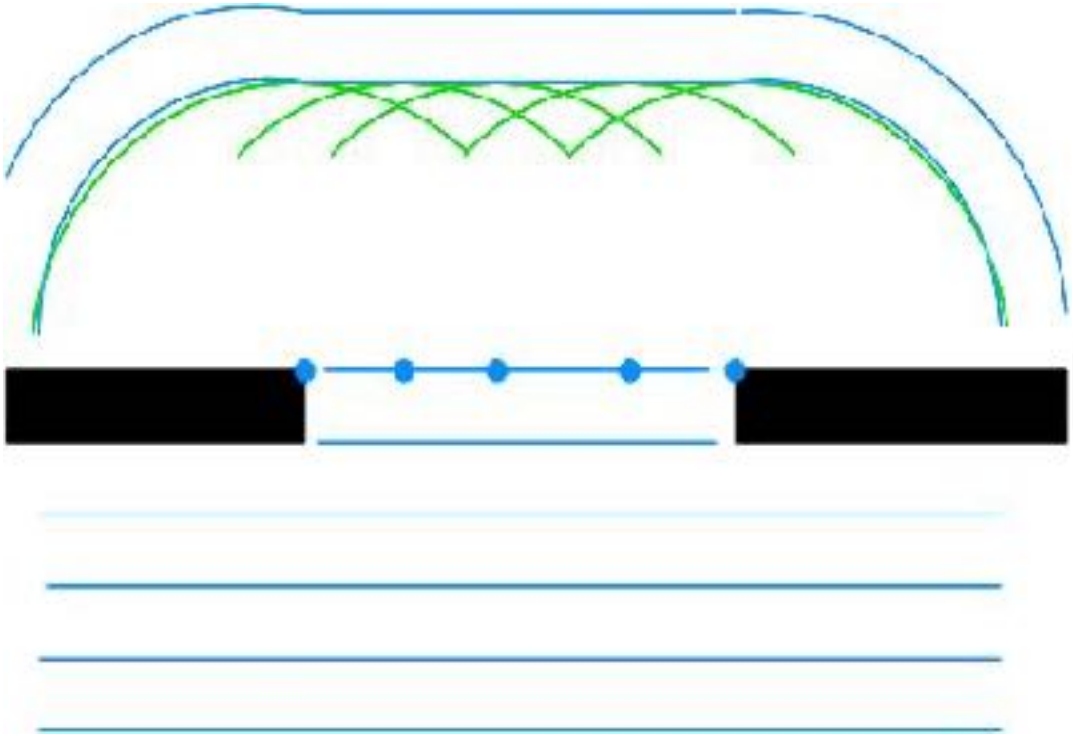
All points on a wavefront are considered sources of tiny wavelets that spread out in the wave direction at the speed of the wave



the new position of the wavefront is a surface tangent to the the wavelets

Dispersion: Huygens' Principle

How waves spread out in the wave direction after passing a narrow opening

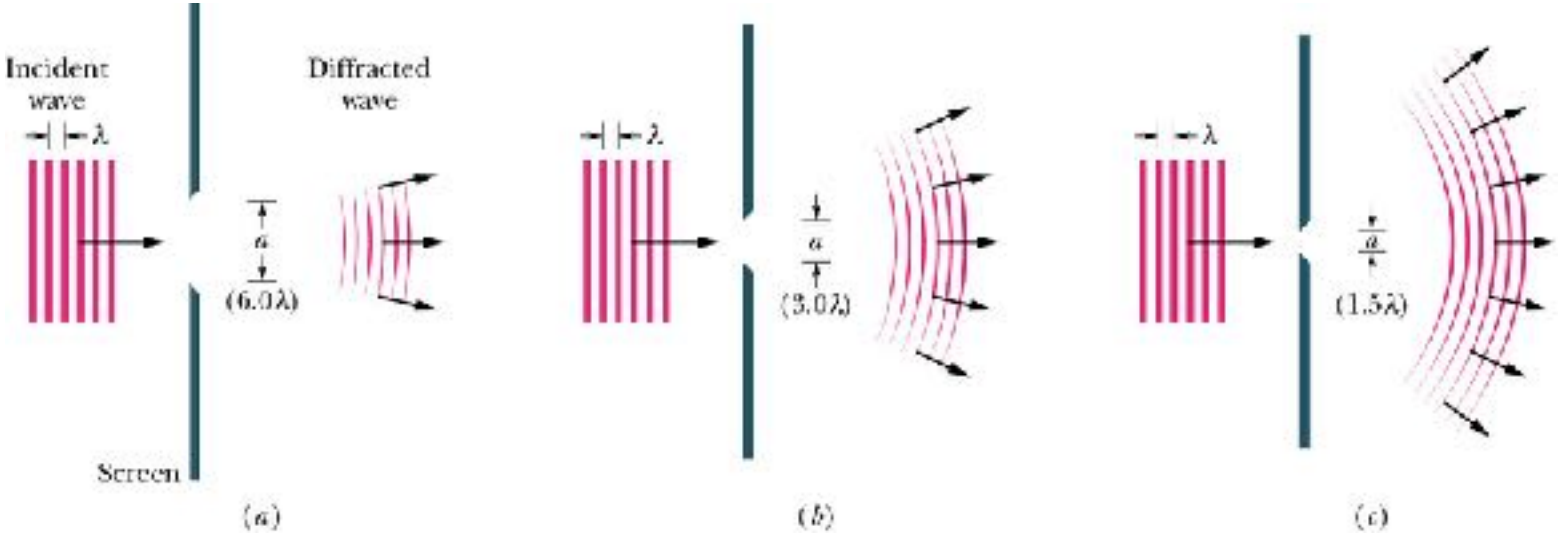


the new position of the wavefront is a surface tangent to the the wavelets

Diffraction

This gives rises to the effect of diffraction

waves going through small apertures tend to spread out, diffract, on the other side.



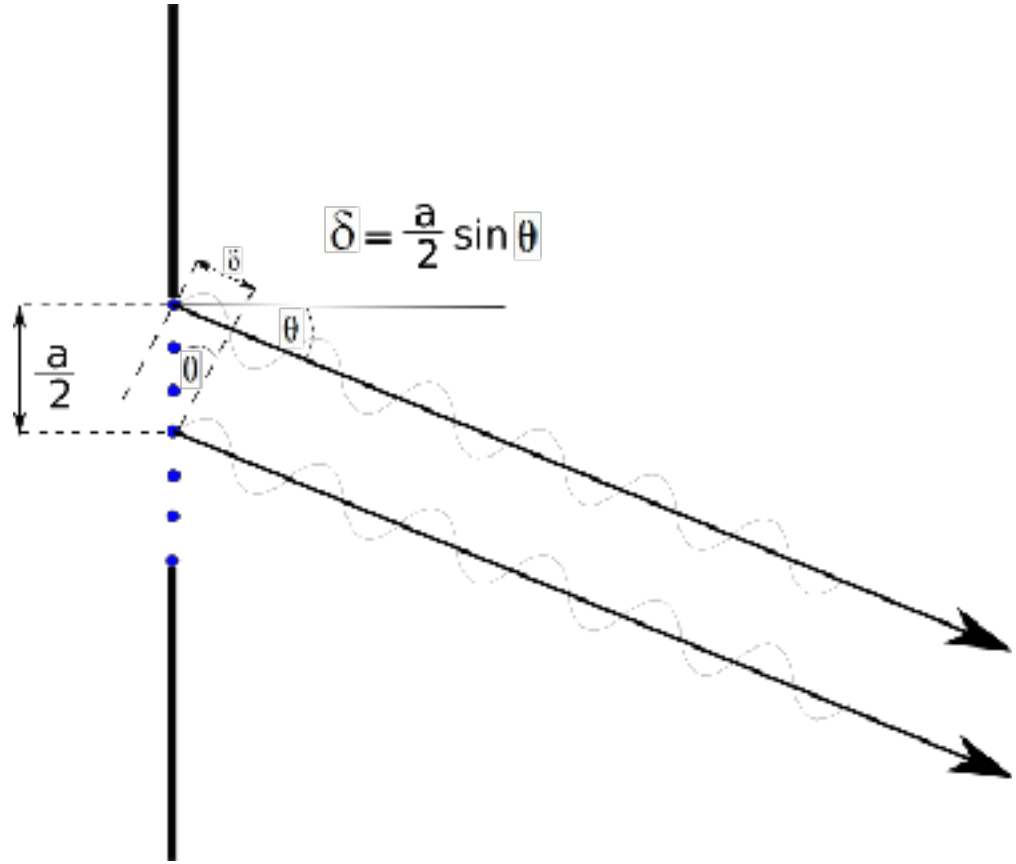
Single-slit diffraction

Imagine a line of Huygens point sources along a slit. Destructive interference occurs when the path-length Difference, $\delta = \lambda/2$

This happens when the rays diverge at specific angle, θ

$$\delta = \frac{a}{2}\sin\theta = \frac{\lambda}{2}$$

$$\boxed{a\ sin\theta = \lambda}$$



Heisenberg Uncertainty Principle

Now recall particles have specific wavelengths related to their momentum

$$\lambda = \frac{h}{p}$$

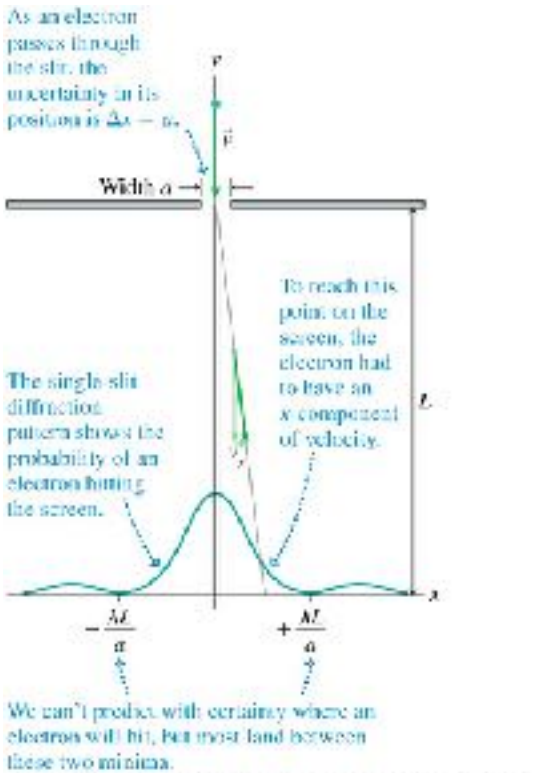
λ = wavelength
h = Plancks'Constant
p = momentum

Because of this if we can locate a particle precisely, we cannot know it's Momentum. Or if we know precisely where a particle is, we cannot know where it is going!

To see why, imagine a situation where We have electrons passing through a Narrow slit, of width a.

Since the electron has a wavelength, λ, This gives rise to a diffraction pattern.

This in turn implies it has some velocity In the x-direction



Heisenberg Uncertainty Principle

So the horizontal velocity v_x is

$$v_x = \frac{distance}{time} = \frac{\frac{\lambda L}{a}}{\frac{L}{v_y}} = \lambda \frac{v_y}{a}$$

Now we substitute the DeBroglie wavelength for the electron

$$v_x = \lambda \frac{v_y}{a} = \frac{h}{mv_y}\frac{v_y}{a} = \frac{h}{ma}$$

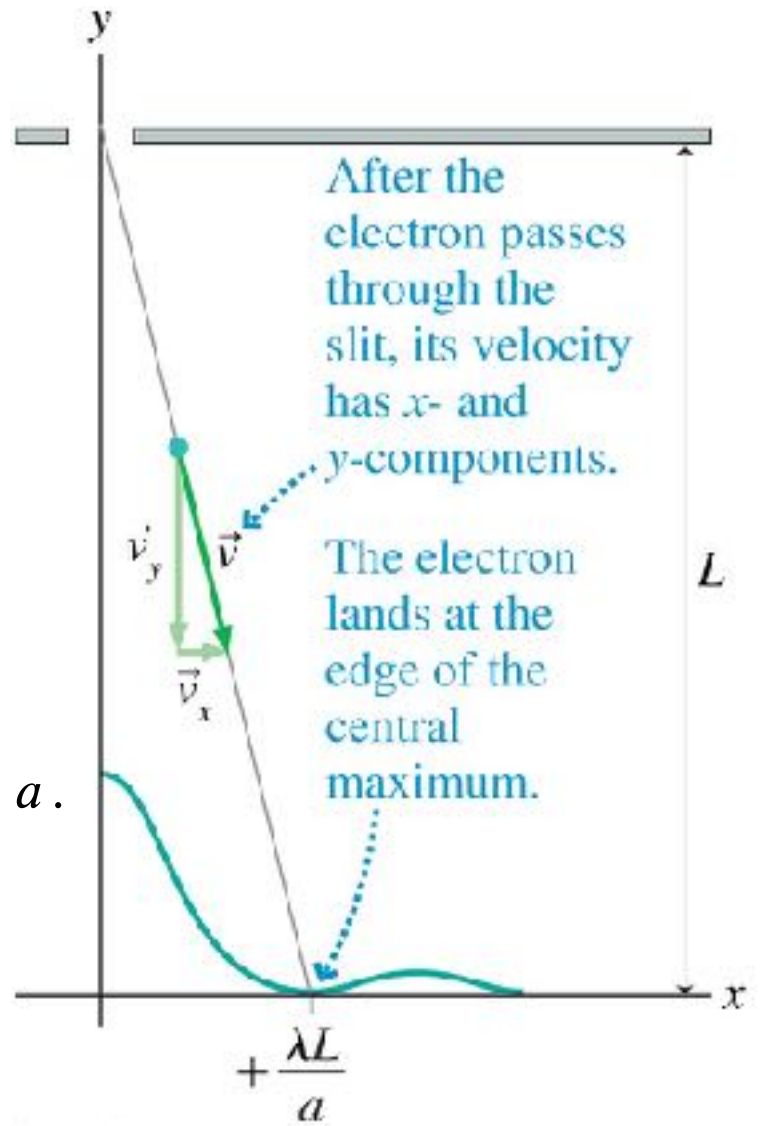
So the x-velocity is uncertain by an amount $\frac{h}{ma}$. Now we've located the position in x to be within a.

So $\Delta x = a$. $\Delta v_x \cong v_x$ so:

$$\Delta x \, m\Delta v_x \cong h$$

And hence

$$\Delta x \, \Delta p_x \cong h$$



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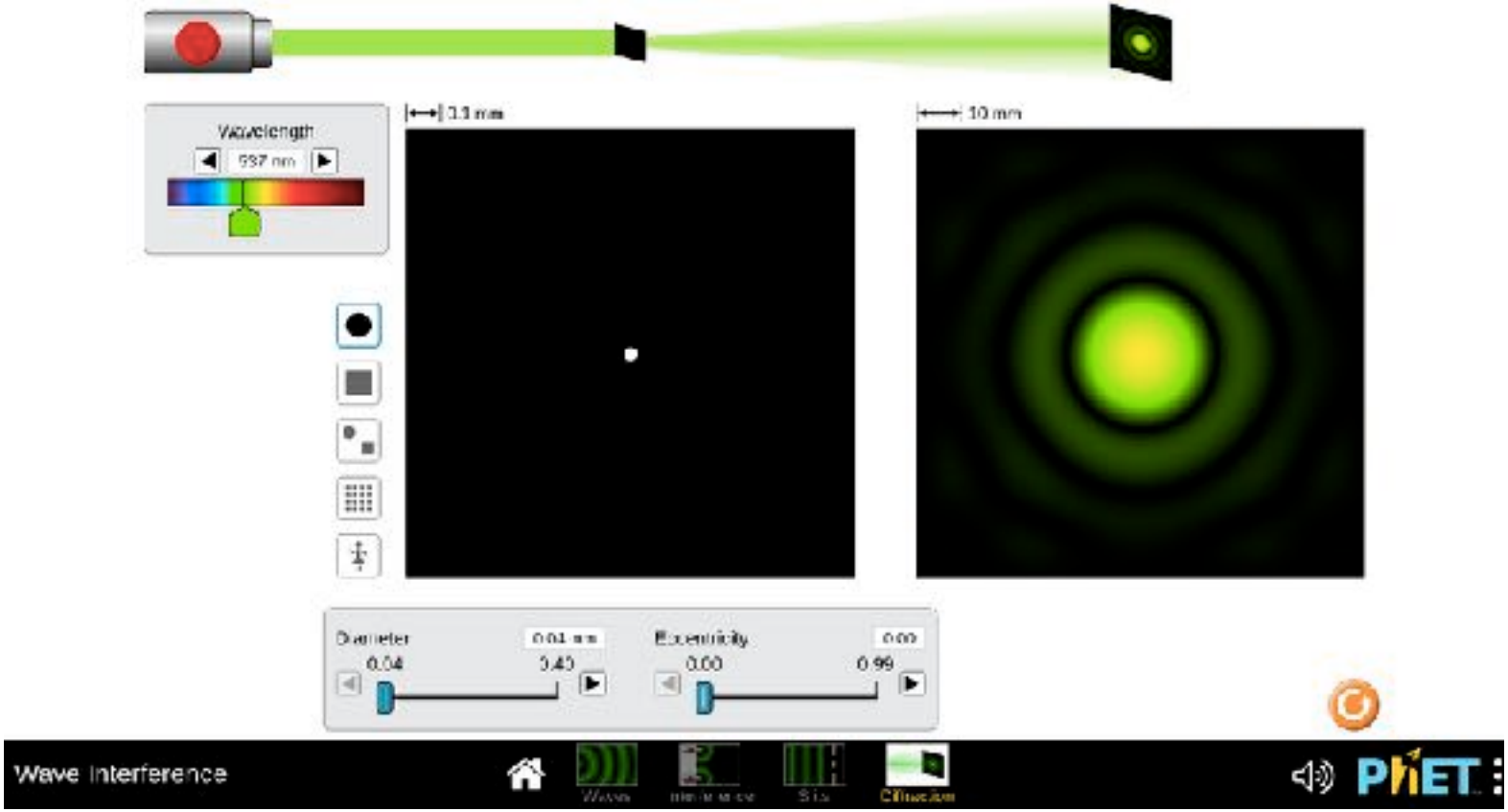
Simulation of Diffraction



https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_en.html

Heisenberg Uncertainty Principle

The German Physicist Werner Heisenberg, using similar arguments showed the most general case is that:

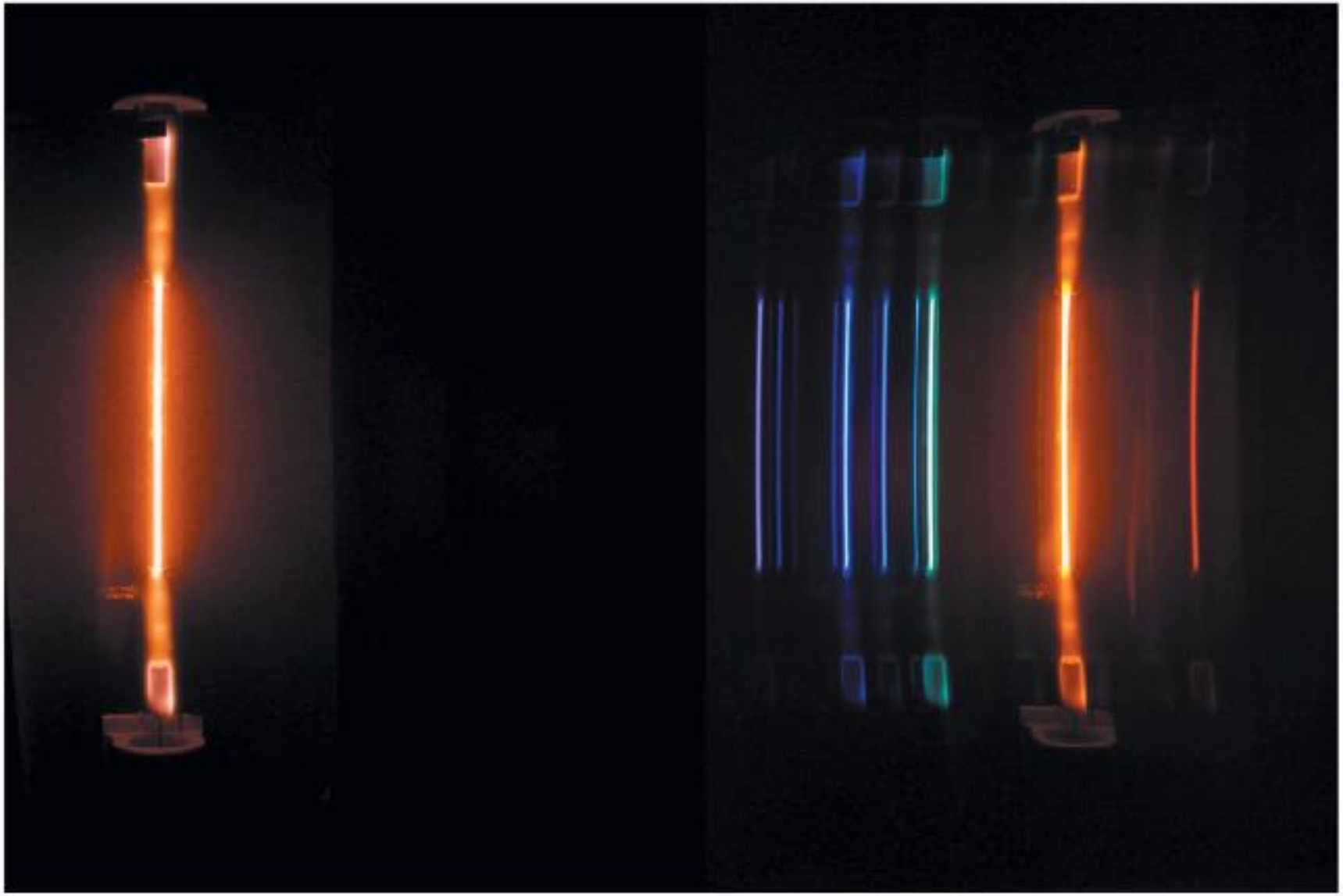
$$\Delta x \Delta p_x \geq \frac{h}{2\pi}$$

He also showed a similar expression holds for Energy and time

$$\Delta t \Delta E \geq \frac{h}{2\pi}$$

ie. If know the precisely the time at which a particle exists, you cannot know its energy. Or a particle may “borrow” some energy, E but it must relinquish it after time $\frac{h}{2\pi E}$.

Hydrogen



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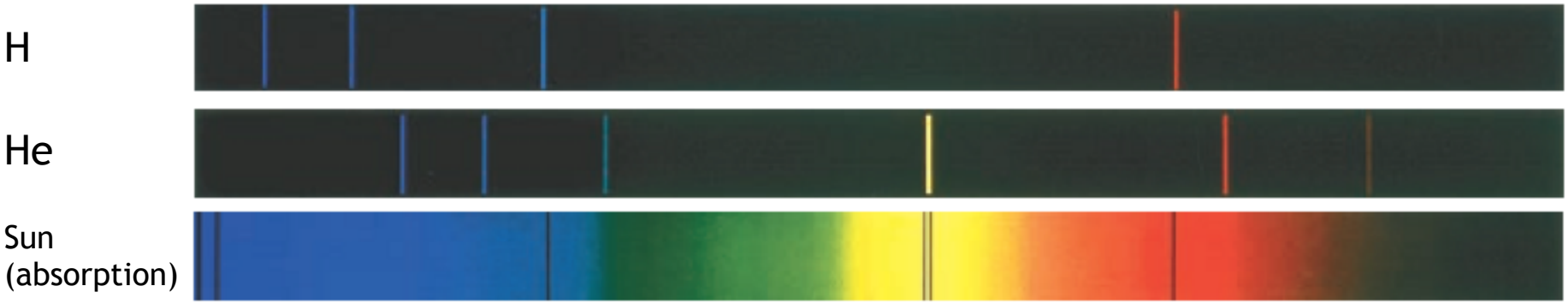
27.11 Quantum matter: early evidence

- 19th century: atoms emit and absorb light at distinct discrete wavelengths
- Different atoms, different spectra



- 1885 JJ Balmer found four lines $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right) \quad n = 3,4,5,\ldots$
- $R = 1.0974 \times 10^7 \mathrm{m}^{-1}$ is the Rydberg constant
- Later, T Lyman $\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$ $n = 2,3\ldots$ and F Paschen $\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$ $n = 4,5,\ldots$
- Clearly something going on here!

27.12 Hydrogen Atom

It is possible to understand the spectra of Hydrogen atoms using
The simple ideas of standing waves and electrostatic attraction.

A Hydrogen atom consists of a single electron orbiting a single proton

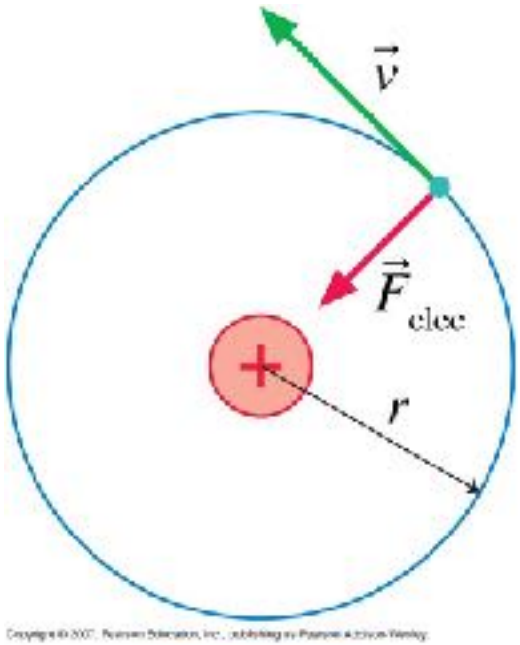
We start by assuming the electrostatic attraction between the electron and Proton is balanced by the centripetal acceleration of the electrons circular Motion.

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2}$$

$$\Rightarrow m^2v^2 = \frac{me^2}{4\pi r\varepsilon_0}$$

$$\Rightarrow p^2 = \frac{me^2}{4\pi r\varepsilon_0}$$

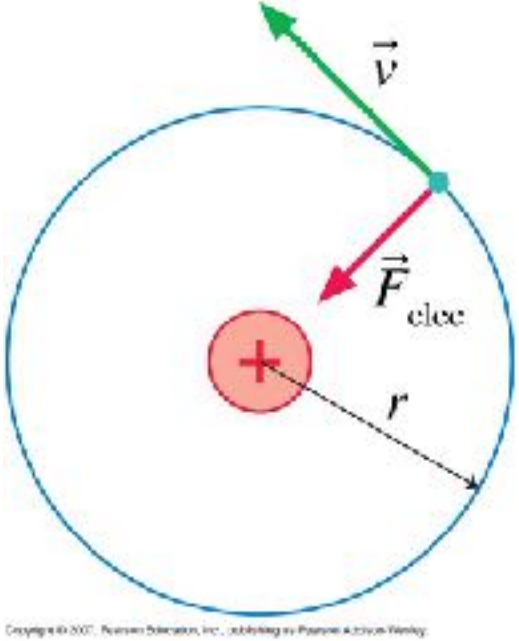
m = electron mass
r = orbital radius
v = electron velocity
e = charge on electron
p = momentum
ε_0 = permittivity



Hydrogen Atom

$$p^2 = \frac{me^2}{4\pi r\varepsilon_0}$$

Now substitute the DeBroglie Wavelength for the electron momentum

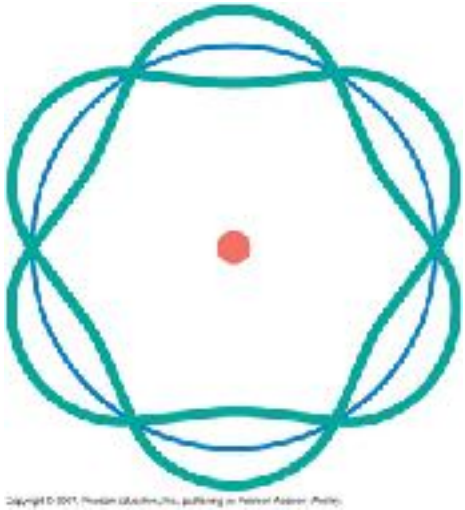


$$\Rightarrow (\frac{h}{\lambda})^2 = \frac{me^2}{4\pi r\varepsilon_0}$$

Rearrange to get λ^2

$$\Rightarrow \lambda^2 = \frac{h^2 4\pi r\epsilon_0}{me^2}$$

Now assume the electron Forms a standing wave Around the proton



ie there is an integral multiple of λ around the proton

m = electron mass
r = orbital radius
v = electron velocity
e = charge on electron
p = momentum
ε_0 = permittivity
h = Plancks' Constant
λ = DeBroglie Wavelength

27.13 Electron de Broglie wavelength

- Circumference $2\pi r = n\lambda$ and $\lambda = \dfrac{h}{p} \Rightarrow p = \dfrac{nh}{2\pi r}$

$$p = 4.5\frac{h}{2\pi r}$$

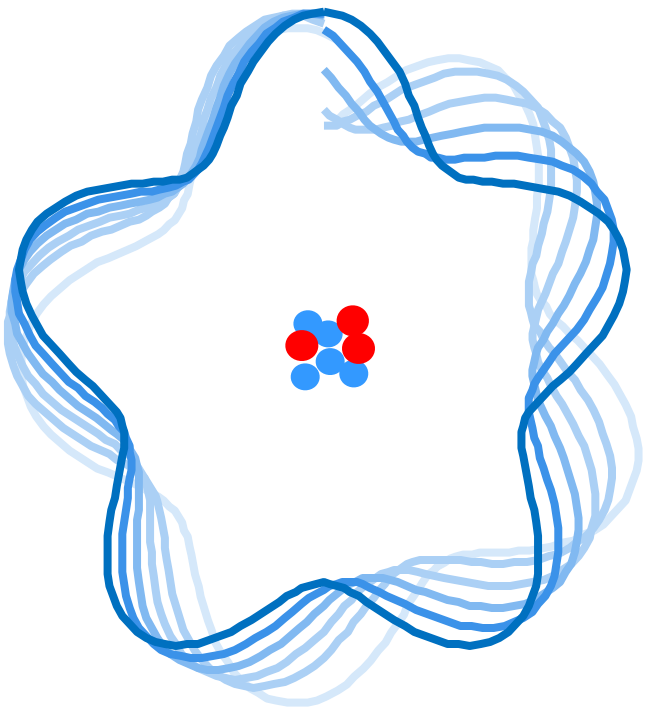
$$p = 4.6\frac{h}{2\pi r}$$

$$p = 4.7\frac{h}{2\pi r}$$

$$p = 4.8\frac{h}{2\pi r}$$

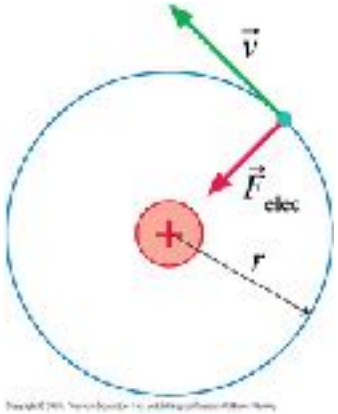
$$p = 4.9\frac{h}{2\pi r}$$

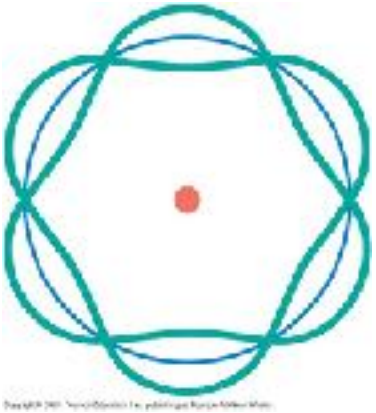
$$p = 5\frac{h}{2\pi r}$$



Hydrogen Atom

Wavelength at radius r $$\lambda^2 = \frac{h^2 4\pi r \epsilon_0}{me^2}$$





Integral multiple of wavelengths requires

$$n\lambda = 2\pi r \qquad \rightarrow \qquad \lambda = \frac{2\pi r}{n}$$

Now substitute this back into our equation for λ^2

$$\left(\frac{2\pi r}{n}\right)^2 = \frac{h^2 4\pi r \epsilon_0}{me^2}$$

Solving for r gives

$$r_n = n^2 \frac{h^2 \epsilon_0}{m\pi e^2}$$

Hydrogen Atom

$$r_n = n^2 \frac{h^2 \epsilon_0}{m \pi e^2}$$

So electrons can only exist at specific radii given by the formula above

Define the "*Bohr Radius*" α_B to be the radius of the inner most electron orbital

$$\alpha_B = r_1 = \frac{h^2 \epsilon_0}{m \pi e^2} = 5.29 \times 10^{-11} \text{ m} = 0.0529 \text{ nm}$$

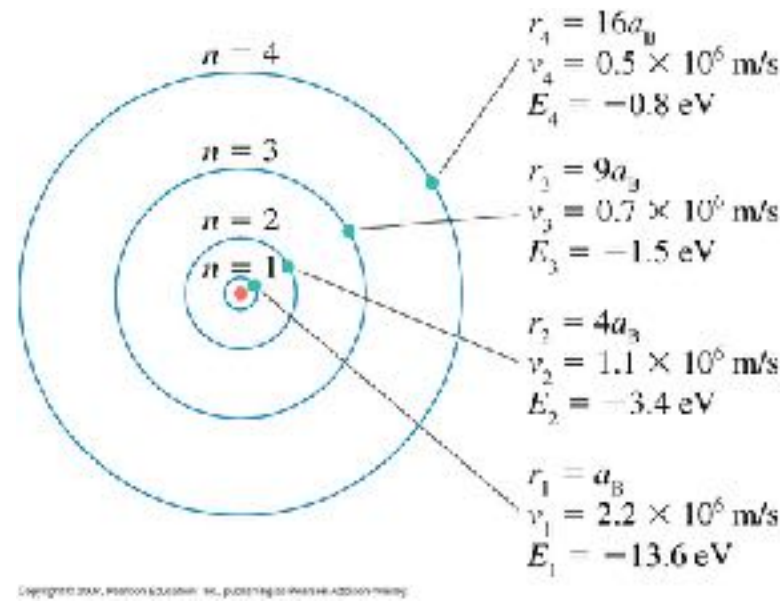
Given the allowed radii we can calculate the energy levels of the Hydrogen Atom

The energy of the electron is the sum of it's Kinetic Energy and Potential Energy

$$E_n = \frac{1}{2} m v_n^2 - \frac{e^2}{4\pi\varepsilon_0 r_n} = \frac{e^2}{8\pi\varepsilon_0 r_n} - \frac{e^2}{4\pi\varepsilon_0 r_n}$$

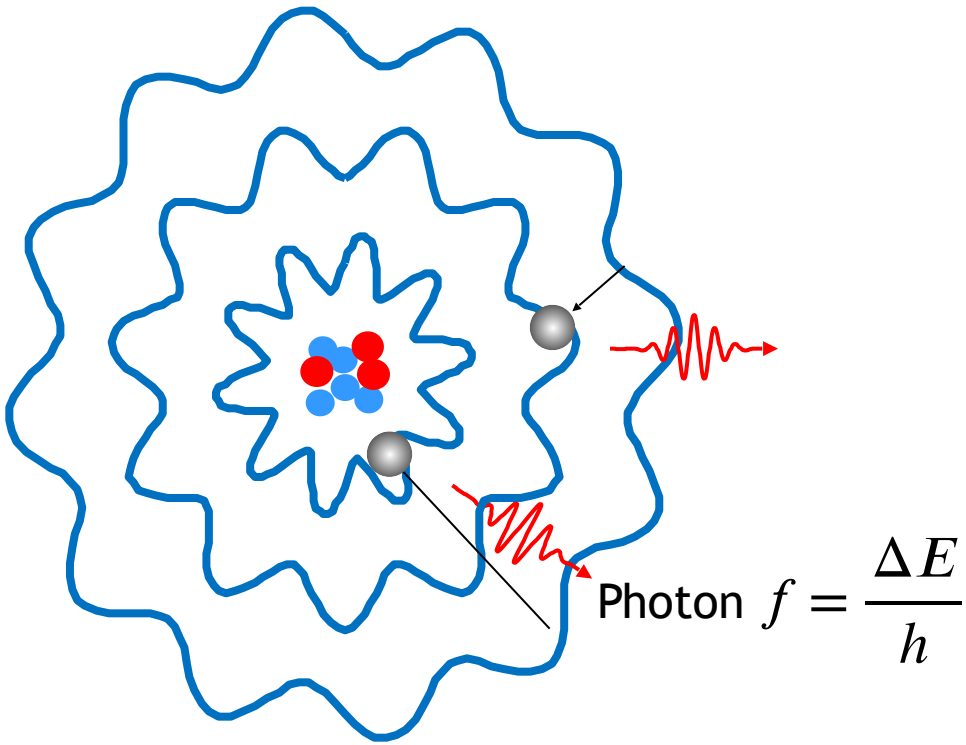
$$= -\frac{1}{n^2}\left(\frac{1}{4\pi\varepsilon_0}\frac{e^2}{2a_B}\right)$$

$$E_n = -\frac{13.60}{n^2} \text{ eV}$$



Discrete orbitals

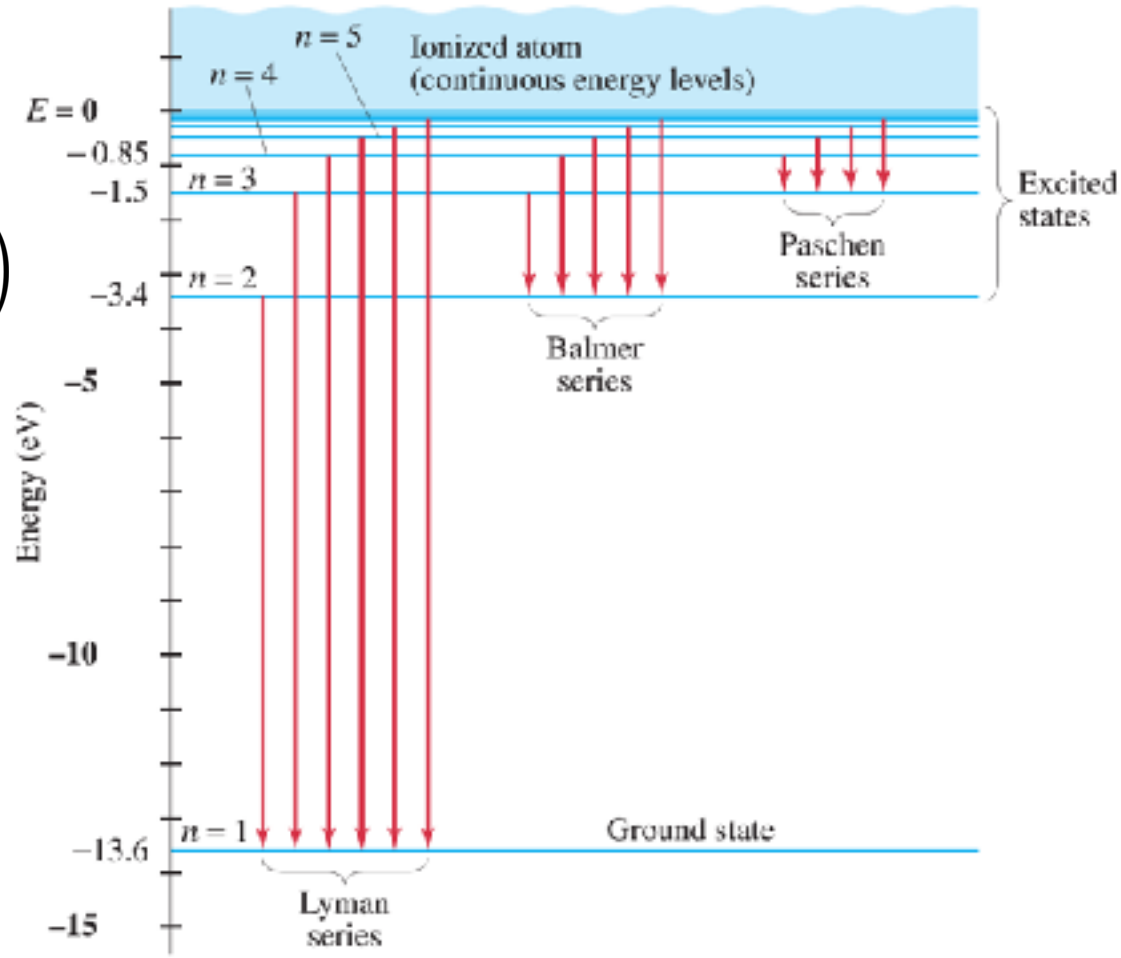
- Integral multiple of λ around the nucleus



Hydrogen energy levels and spectra

- Each level connected to many others
- Lyman series = excited states $n > 1$ to ground state, $m = 1$
- Balmer series = first excited state, $m = 2$

$$f_{nm} = \frac{E_n - E_m}{h} = \frac{13.6\text{eV}}{h}\left(\frac{1}{m^2} - \frac{1}{n^2}\right)$$



Of-3 Spectral Sources

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Transitions in Hydrogen

Which of the following electron transitions between two energy states in the hydrogen atom corresponds to the emission of a photon with the longest wavelength?

(a) 2 -> 5

(b) 5 -> 2

(c) 5 -> 8

(d) 8 -> 5

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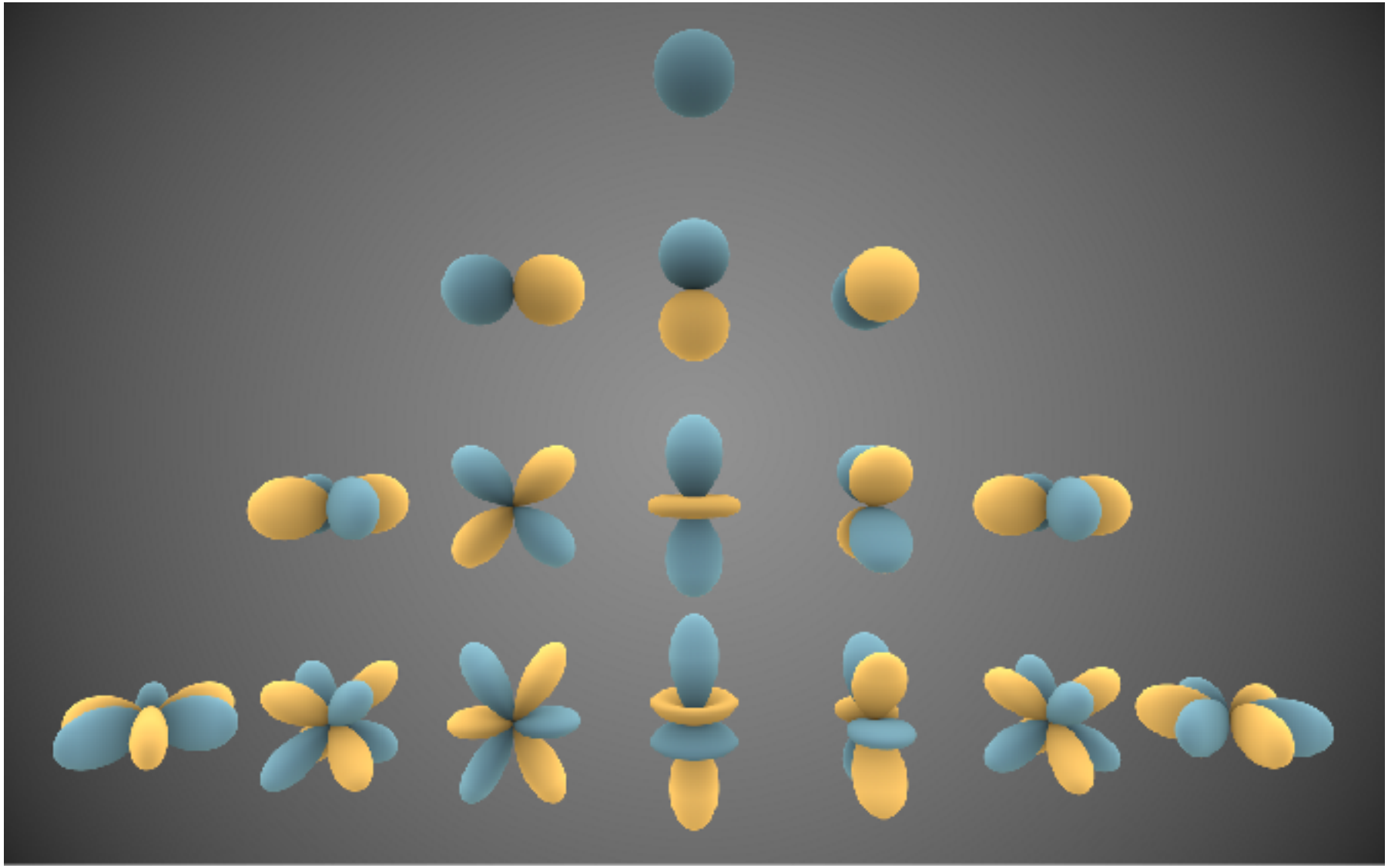
Hydrogen: p + e = simple?

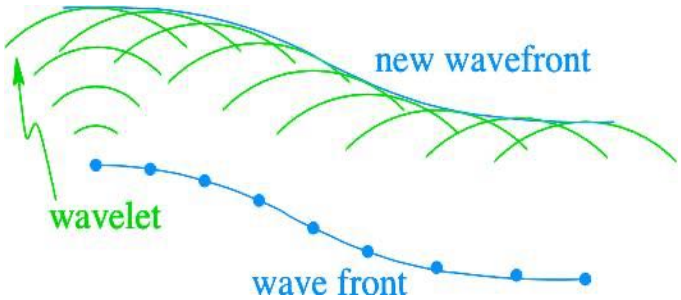


Summary

DeBroglie Wavelength

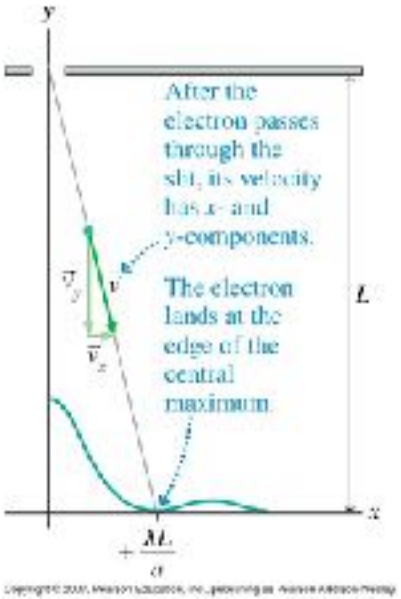
$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Huygens' Principle



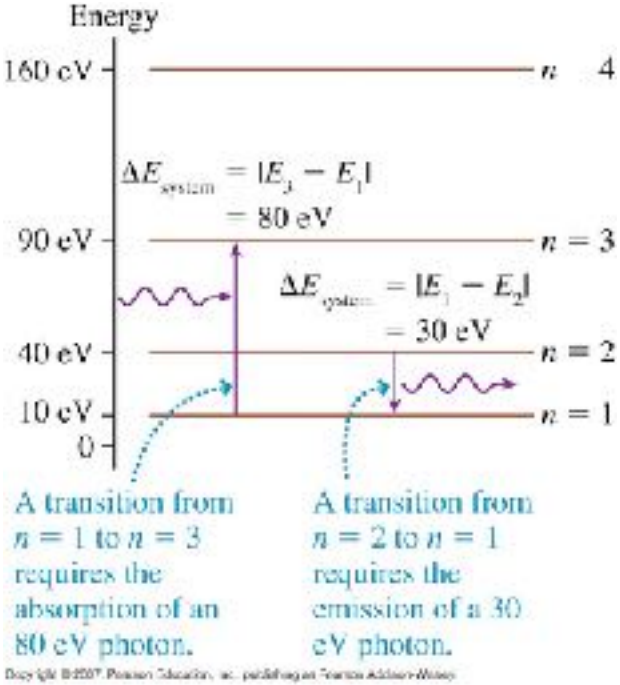
the new position of the wavefront is a surface tangent to the the wavelets

Diffraction



Uncertainty Principle

$$\Delta x \Delta p_x \geq \frac{h}{2\pi}$$

$$\Delta t \Delta E \geq \frac{h}{2\pi}$$

Energy is Quantised